# Grade 3
# Ontario Curriculum
# Spatial Sense

Fencing the Farm — Grade 3 Measurement

# Unit Overview: Fencing the Farm — Grade 3 Measurement

**Grade:** Grade 3 | Strand: Spatial Sense

**Target Expectations:** E2.1, E2.2, E2.3, E2.4, E2.5, E2.6, E2.7, E2.8, E2.9

**Learning Goal:** Children will measure perimeter, area, capacity, mass, and time using appropriate units and tools.

| Lesson | Lesson Title | Learning Goal |
|---|---|---|
| Day 1 | **Fence the Farm (Perimeter)** | I can measure perimeter and pick the right metric unit. |
| Day 2 | **Compare Areas** | I can compare areas of shapes by matching or covering. |
| Day 3 | **Square Centimetres** | I can measure area in square centimetres and square metres. |
| Day 4 | **Capacity and Mass** | I can measure capacity and mass with non-standard units. |
| Day 5 | **Time and Big Farm Capstone** | I can tell time and pick the right measurement tool. |

# Lesson 1: Fence the Farm (Perimeter)

**Goal:** PERIMETER is the distance AROUND a shape. Pick mm, cm, m, or km depending on size.
**Expectation:** E2.1, E2.2

**Learning Goal:** "I can measure perimeter and pick the right metric unit."

**Materials:** M1_polygon_set, M2_unit_chart, M7_measurement_chart, M10_unit_poster

## 1. Minds On (12 mins): Roo's Farm Fence

Roo the Ruler Rabbit: Farmer Phil needs FENCES around 4 fields. The PERIMETER tells him how much fence to buy! Roo opens: 'WALK AROUND, ADD the SIDES — that's PERIMETER!'

Bring children to the carpet. Roo the Ruler Rabbit puppet. 'Measurement friends — last year we measured LENGTHS in cm and m. This year we measure all the way AROUND a shape — that's PERIMETER!'

## 2. Action (25 mins): Measure 4 Perimeters

- Hand out Polygon Cards + ruler.
- For each, write the unit (mm/cm/m/km).
- Stretch: construct a rectangle with perimeter 20 cm.
- Help with addition strategy — column or in-pairs.
- Circulate to record diagnostic.
- INVENT-AND-COMPARE: Pairs find perimeter of a hexagon (6 sides of 2 cm) ANY way.

## 3. Consolidation (13 mins): Roo's Fence Brag

Discuss:
- What's PERIMETER?
- Which unit for a bug?
- How many cm in 1 m?
- Same perimeter — different shape?
- Roo's preview: tomorrow we COMPARE AREAS (Lesson 2 — E2.1).

Exit: Roo's perimeter chant — 'WALK AROUND, ADD the SIDES — that's PERIMETER!' (3x).

# Lesson 2: Compare Areas

**Goal:** AREA is the SPACE INSIDE a shape. Compare by MATCHING, COVERING, or DECOMPOSING.
**Expectation:** E2.7, E2.8

**Learning Goal:** "I can compare areas of shapes by matching or covering."

**Materials:** M3_tile_set, M7_measurement_chart, M10_unit_poster

## 1. Minds On (12 mins): Roo's Tile Cover

Roo: AREA is HOW MUCH space is INSIDE. Cover the field with tiles to compare! Roo opens: 'ROWS times COLUMNS — that's AREA!' Yesterday Roo did the BOUNDARY. Today we cover the INSIDE.

Bring children to the carpet. Roo the Ruler Rabbit puppet. 'Measurement friends — yesterday we measured AROUND. Today we measure INSIDE — that's AREA!'

## 2. Action (25 mins): Cover and Compare 4 Shapes

- Hand out shape outlines + 30 tiles.
- Compare 2 pairs of shapes — which has bigger area?
- Stretch: decompose 1 shape into smaller parts and recompose.
- Help with edge-to-edge tile placement.
- Circulate to record formative.
- STRATEGY COMPARE: Two pairs share their area methods — Pair A counts one-by-one; Pair B uses rows × columns.

## 3. Consolidation (13 mins): Roo's Cover Brag

Discuss:
- What's AREA?
- How is area DIFFERENT from perimeter?
- What if tiles OVERLAP?
- Can 2 shapes have the same area?

Exit: Roo's area chant — 'Roo says: ROWS times COLUMNS — that's AREA!' (3x).

# Lesson 3: Square Centimetres

**Goal:** AREA is measured in SQUARE UNITS — $cm^2$ for small, $m^2$ for big.

**Expectation:** E2.9

**Learning Goal:** "I can measure area in square centimetres and square metres."
**Materials:** M3_tile_set, M9_grid_paper, M7_measurement_chart, M10_unit_poster

### 1. Minds On (12 mins): Roo's Square Centimetres

Roo: AREA needs a UNIT. We use SQUARE CENTIMETRES ($cm^2$). Count them inside! Roo opens: '$cm^2$ is a TILE — count the TILES!'

Bring children to the carpet. Roo the Ruler Rabbit puppet with grid paper. 'Measurement friends — yesterday we covered with tiles. Today our tile is OFFICIAL: 1 cm × 1 cm = 1 SQUARE CENTIMETRE = $cm^2$.'

### 2. Action (25 mins): Count $cm^2$ for 4 Shapes

- Hand out grid paper + shapes.
- For 2 shapes pick $cm^2$ or $m^2$.
- Stretch: count $cm^2$ for 1 curved shape (whole + half squares).
- Help with grid alignment.
- Circulate to record formative.
- STRATEGY COMPARE: Two pairs share L-shape area — one decomposes into two rectangles, one counts tiles.

### 3. Consolidation (13 mins): Roo's Square Brag

Discuss:
- What's a SQUARE CENTIMETRE?
- When use $m^2$ instead of $cm^2$?
- How count for a CIRCLE?
- Why ESTIMATE first?

Exit: Roo's $cm^2$ chant — 'Roo says: $cm^2$ is a TILE — count the TILES!' (3x).

# Lesson 4: Capacity and Mass

**Goal:** CAPACITY = how much it HOLDS. MASS = how HEAVY. Different units = different counts.
**Expectation:** E2.3, E2.4, E2.5

**Learning Goal:** "I can measure capacity and mass with non-standard units."
**Materials:** M4_balance_card, M8_capacity_set, M7_measurement_chart, M10_unit_poster

## 1. Minds On (12 mins): Cal's Pour and Weigh

Cal: CAPACITY tells how much a cup HOLDS. MASS tells how HEAVY a pumpkin is. Pour and weigh! Roo opens: 'CAPACITY for what FITS, MASS for what WEIGHS!'

Bring children to the carpet. Cal the Calendar Cat puppet with stopwatch. 'Measurement friends — today we measure how much HOLDS (capacity) and how HEAVY (mass).'

## 2. Action (25 mins): Capacity and Mass Stations

- Hand out Capacity Cards.
- Hand out Pan Balance Cards.
- Stretch: same pail, count in cups AND in scoops.
- Help with the unit-size relationship.
- Circulate to record formative.
- STRATEGY COMPARE: Pairs predict the cup-to-scoop count for a NEW pail BEFORE pouring, then test.

## 3. Consolidation (13 mins): Cal's Measure Brag

Discuss:
- What's CAPACITY?
- What's MASS?
- Why does cup count differ from scoop count?
- What's an OVERFILL?

Exit: Roo+Cal mass+capacity refrain — 'Roo measures CAPACITY, Cal weighs MASS — same NUMBER, different ATTRIBUTE!' (3x).

# Lesson 5: Time and Big Farm Capstone

**Goal:** TELL TIME in hours, minutes, seconds. Use the right TOOL for each measurement.
**Expectation:** E2.6

**Learning Goal:** "I can tell time and pick the right measurement tool."

**Materials:** M1_polygon_set, M3_tile_set, M4_balance_card, M5_clock_set

### 1. Minds On (12 mins): Cal's Clock Time

Cal: TIME on analog and digital clocks. Today's BIG case — perimeter, area, capacity/mass, and TIME! Roo + Cal open: 'Measure AROUND, measure the FILL, measure the TIME!'

Bring children to the carpet. Roo and Cal puppets in farmer hats. 'Friends — today is our CAPSTONE. Last 4 days we did perimeter, area, area in $cm^2$, capacity and mass. Today we add TIME!'

### 2. Action (25 mins): The Big Farm

- Hand out templates.
- Stations 1-4: perimeter, area, capacity/mass, time.
- Stretch: write a 'farmer's brag' — one sentence per station.
- Help with clock reading at Station 4.
- Circulate to record summative evidence.
- STRATEGY COMPARE at perimeter station: pause at minute 15.

### 3. Consolidation (13 mins): Roo and Cal's Big Farm Brag

Discuss:
- What's PERIMETER vs AREA?
- When use $cm^2$ vs $m^2$?
- What's CAPACITY?
- What's HOURS, MINUTES, SECONDS?

Exit: Roo + Cal capstone refrain — Roo: 'Measure AROUND, measure the FILL!' Cal: 'Measure the TIME!' Class: 'That's MEASUREMENT!' (3x).

Name: ____________________ Date: ____________________

# Roo's Fence Math

**Learning Goal:** I can measure perimeter and pick the right metric unit.

## Mission

1. For 4 polygons, add up all sides to find the perimeter.
2. For 4 objects, circle the BEST unit: mm, cm, m, or km.
3. Draw a rectangle with perimeter 16 cm.

**Part 3: Build a polygon.**

Draw a rectangle with perimeter 16 cm. Label sides.

(3 parts on the printable worksheet)

Name: ____________________ Date: ____________________

# Roo's Tile Compare

**Learning Goal:** I can compare areas of shapes by matching or covering.

## Mission

1. Count tiles inside 4 shapes.
2. For 2 shape pairs, circle BIGGER, SMALLER, or SAME area.
3. Draw 2 DIFFERENT shapes that have the SAME area (each 6 tiles).

**Part 3: Find SAME area.**

Draw 2 DIFFERENT shapes that have the SAME area (each 6 tiles).

(3 parts on the printable worksheet)

Name: ______________________ Date: ______________________

# Roo's Square Cm Hunt

**Learning Goal:** I can measure area in square centimetres and square metres.

**Instructions for the Student:**

1. Count $cm^2$ inside 4 shapes on a grid.
2. For 4 objects, circle the right area unit: $cm^2$ or $m^2$.
3. Estimate the area of the circle on the grid by counting whole + half squares.

**Part 3: Estimate a circle.**

Estimate the area of the circle on the grid by counting whole + half squares.

(4 parts on the printable worksheet)

Name: ______________________ Date: ______________________

# Cal's Pour and Weigh

**Learning Goal:** I can measure capacity and mass with non-standard units.

**Instructions for the Student:**

1. For 4 container pairs, circle which holds MORE.
2. For 4 pan balance images, write HEAVIER, LIGHTER, or EQUAL.
3. If a pail = 6 cups and 1 cup = 2 scoops, how many SCOOPS?

**Part 3: Cups vs scoops.**

If a pail = 6 cups and 1 cup = 2 scoops, how many SCOOPS?

(3 parts on the printable worksheet)

Name: ______________________________ Date: ______________________________

# Roo and Cal's Big Farm

**Learning Goal:** I can tell time and pick the right measurement tool.

**Instructions for the Student:**

1. For 2 shapes, find PERIMETER (cm) and AREA ($cm^2$).
2. Compare 1 capacity pair and 1 mass pair.
3. Read 4 clocks (mix of analog + digital).

**Part 2: Capacity and mass.**

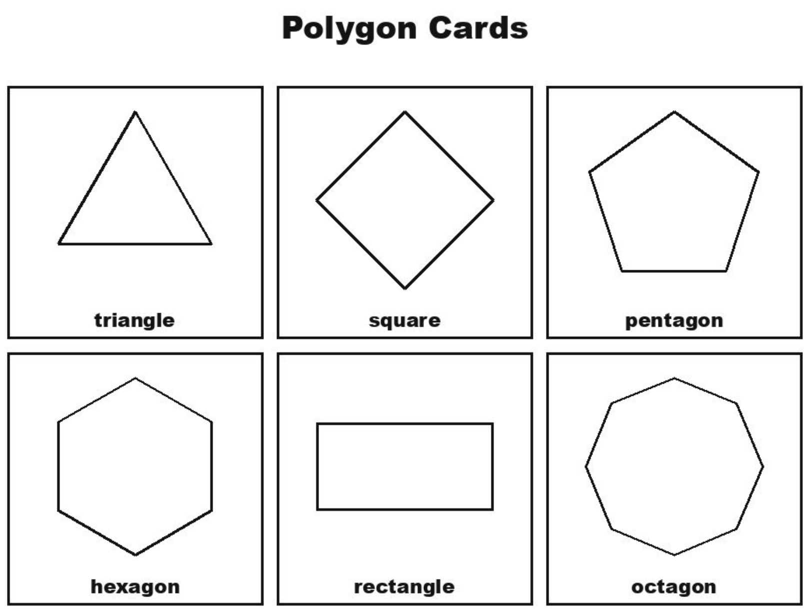
Compare 1 capacity pair and 1 mass pair.

(4 parts on the printable worksheet)

# Polygon Cards Set

(M1_polygon_set)

12 paper polygons with side lengths labelled.




PRINT SPECS
- Size: letter portrait
- Colour: bw
- Pages per set: 1
- Laminate: yes

USE IN CLASS
- Quantity: 10 sets
- Lessons: 1, 5
- Prep: ~25 min

TEACHER PREP
1. Print 10 sets.
2. Laminate.
3. Cut into 12 cards per set.
4. Store in labelled bag.

# Metric Unit Chart

(M2_unit_chart)

Anchor chart with mm, cm, m, km benchmarks.

## Benchmarks

| | | | |
|---|---|---|---|
| **1 cm** | fingernail | **10 cm** | thumb to pinky |
| **1 m** | long step | **1 g** | paper clip |
| **100 g** | stick of butter | **1 kg** | 1 L of water |
| **1 mL** | drop | **1 L** | milk carton |

PRINT SPECS
Size: letter portrait
Colour: bw
Pages per set: 1
Laminate: yes

USE IN CLASS
Quantity: 1 chart
Lessons: 1, 5
Prep: ~10 min

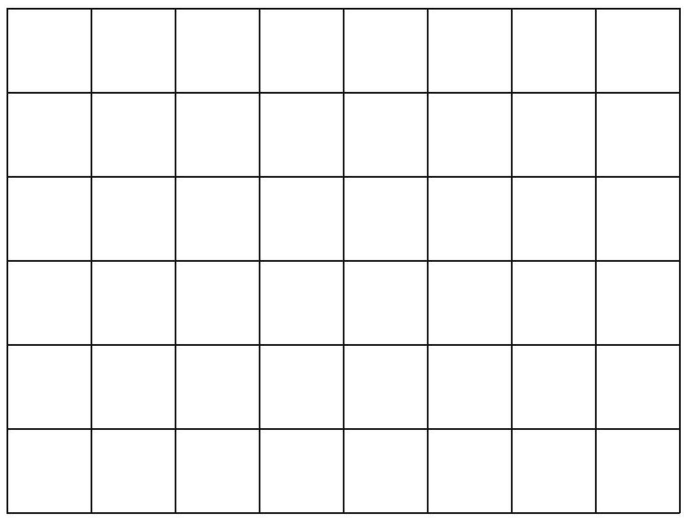
TEACHER PREP
1. Print 1 copy.
2. Laminate.
3. Tape to front wall before Lesson 1.

# Square Tiles Set

(M3_tile_set)

60 paper square tiles (1 $cm^2$ each).

**Area Grid — 8 × 6**



PRINT SPECS
- Size: letter portrait
- Colour: bw
- Pages per set: 2
- Laminate: yes

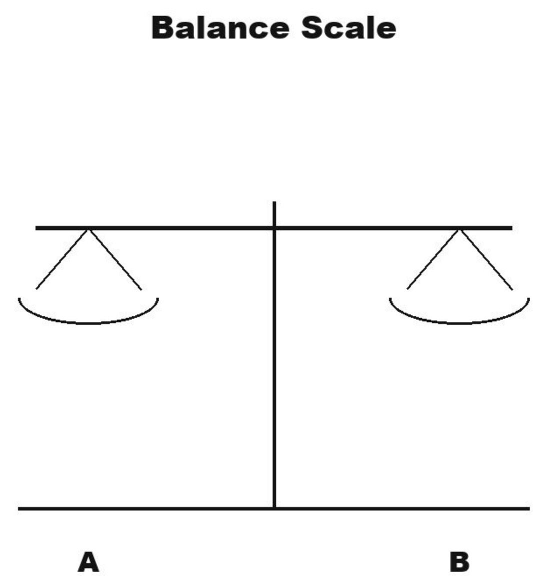
USE IN CLASS
- Quantity: 5 sets
- Lessons: 2, 3, 5
- Prep: ~30 min

TEACHER PREP
1. Print 5 sets.
2. Laminate.
3. Cut into 60 tiles per set.
4. Store in labelled bag.

# Pan Balance Cards

(M4_balance_card)

12 paper pan balance cards with weighable items.




PRINT SPECS
- Size: letter portrait
- Colour: bw
- Pages per set: 1
- Laminate: yes

USE IN CLASS
- Quantity: 10 sets
- Lessons: 4, 5
- Prep: ~20 min

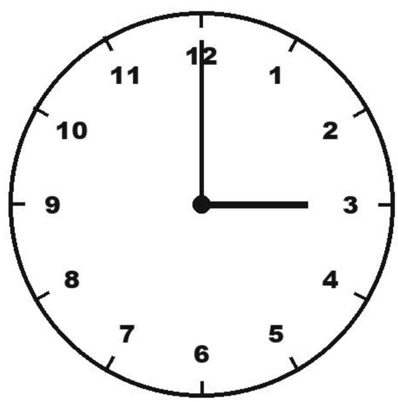
TEACHER PREP
1. Print 10 sets.
2. Laminate.
3. Cut into 12 cards per set.
4. Store in labelled bag.

# Analog and Digital Clock Cards

(M5_clock_set)

16 clock cards mixing analog and digital.

**Clock — 03:00**




PRINT SPECS
- Size: letter portrait
- Colour: bw
- Pages per set: 1
- Laminate: yes

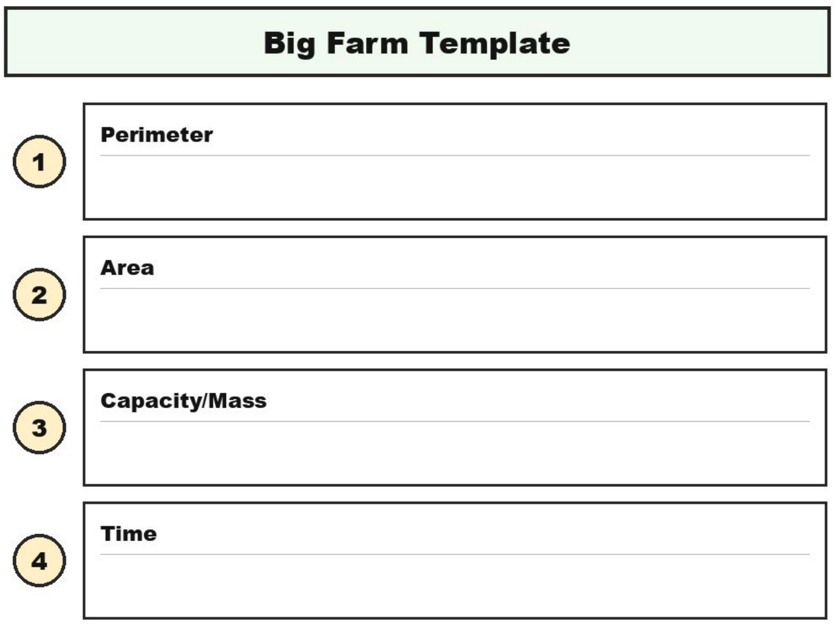
USE IN CLASS
- Quantity: 10 sets
- Lessons: 5
- Prep: ~25 min

TEACHER PREP
1. Print 10 sets.
2. Laminate.
3. Cut into 16 cards per set.
4. Store in labelled bag.

# Big Farm Template

(M6_capstone_template)

Letter-size capstone template with 4 sections.




PRINT SPECS
Size: letter portrait
Colour: bw
Pages per set: 1
Laminate: no

USE IN CLASS
Quantity: 25 templates
Lessons: 5
Prep: ~5 min

TEACHER PREP
1. Print 25 templates.
2. Stack near teacher table.
3. No lamination.

# Measurement Vocabulary Chart

(M7_measurement_chart)

Anchor chart with G3 measurement vocabulary.

**Measurement Vocabulary Chart**

| Perimeter | Capacity |
| --- | --- |
| • Area<br>• Millimetre<br>• Kilometre | • Mass<br>• Square Centimetre<br>• Square Metre |

PRINT SPECS
Size: letter portrait
Colour: bw
Pages per set: 1
Laminate: yes

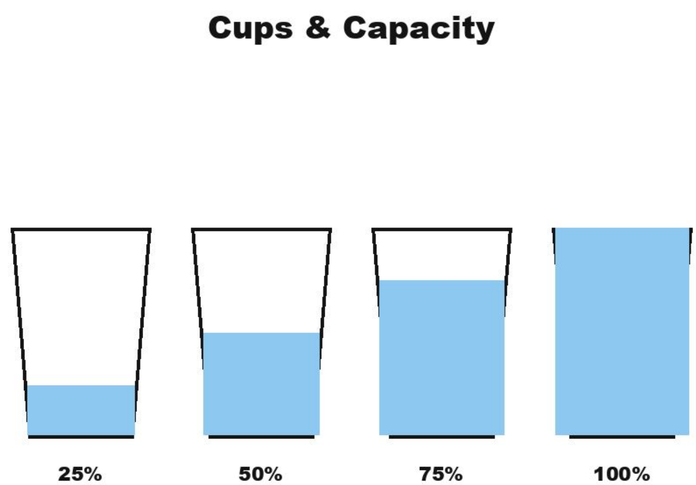
USE IN CLASS
Quantity: 1 chart
Lessons: 1, 2, 3, 4, 5
Prep: ~10 min

TEACHER PREP
1. Print 1 copy.
2. Laminate.
3. Tape to front wall before Lesson 1.

# Capacity Cup Cards

(M8_capacity_set)

20 paper container cards with cup/scoop unit counts.




PRINT SPECS
- Size: letter portrait
- Colour: bw
- Pages per set: 1
- Laminate: yes

USE IN CLASS
- Quantity: 10 sets
- Lessons: 4, 5
- Prep: ~25 min

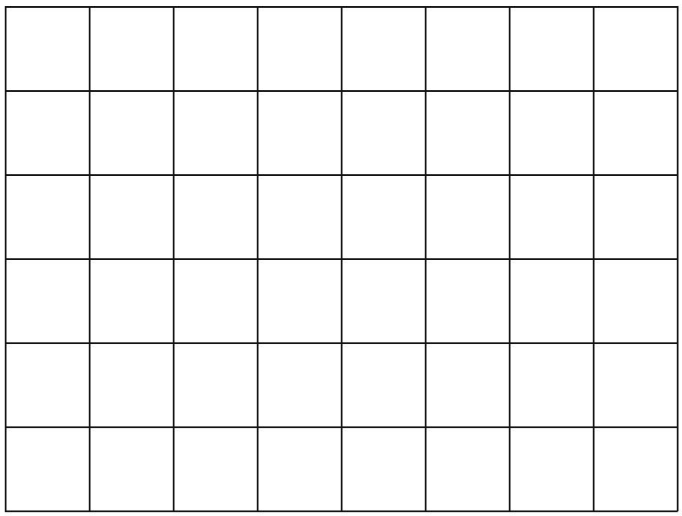
TEACHER PREP
1. Print 10 sets.
2. Laminate.
3. Cut into 20 cards per set.
4. Store in labelled bag.

# Cm$^2$ Grid Paper

(M9_grid_paper)

Letter-size grid paper with 1 cm × 1 cm squares.

**Area Grid — 8 × 6**



PRINT SPECS
- Size: letter portrait
- Colour: bw
- Pages per set: 1
- Laminate: yes

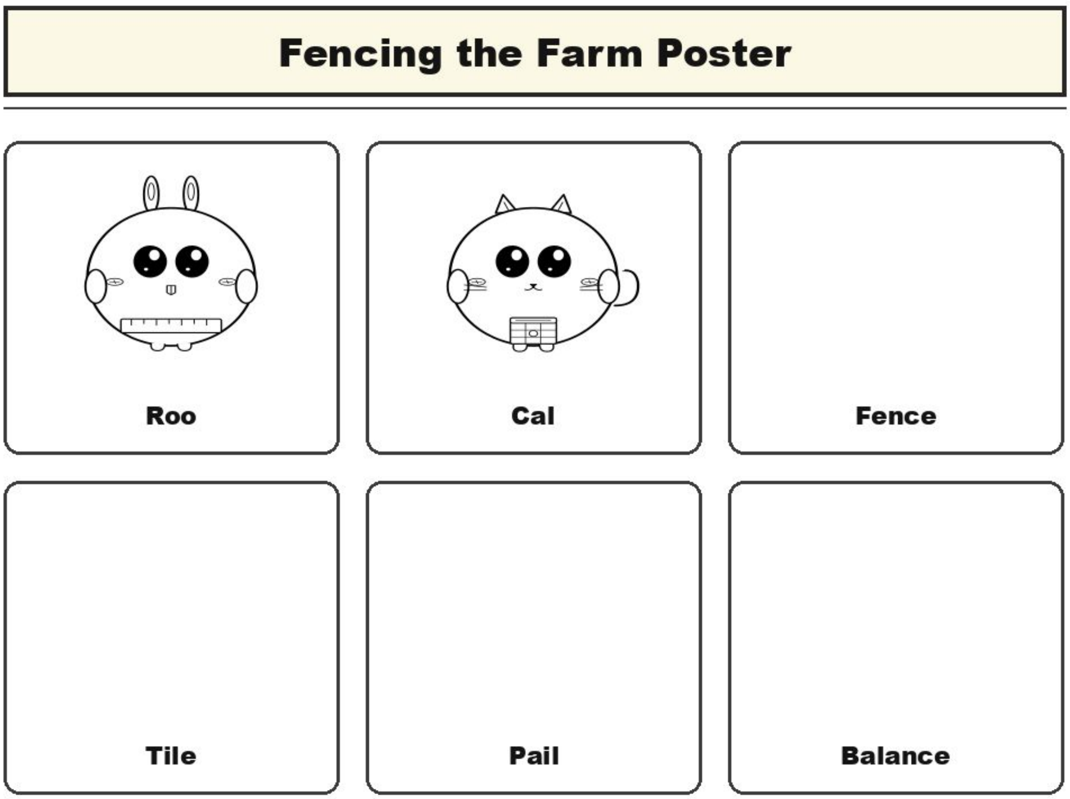
USE IN CLASS
- Quantity: 10-12 sheets
- Lessons: 3, 5
- Prep: ~10 min

TEACHER PREP
1. Print 12 sheets.
2. Laminate (optional).
3. Distribute one per pair.

# Fencing the Farm Poster

(M10_unit_poster)

Unit anchor poster of Roo and Cal at a farm.




PRINT SPECS
Size: letter landscape
Colour: bw
Pages per set: 1
Laminate: yes

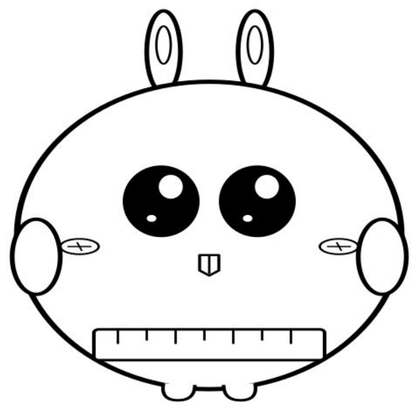
USE IN CLASS
Quantity: 1 poster
Lessons: 1, 2, 3, 4, 5
Prep: ~10 min

TEACHER PREP
1. Print 1 copy on letter or 11x17.
2. Laminate.
3. Tape to front wall on Day 1.

# Roo Puppet

(char_roo_puppet)

Paper puppet of Roo the Ruler Rabbit for teacher use in all 5 lessons.



**Roo the Ruler Rabbit**

PRINT SPECS
Size: letter portrait
Colour: bw
Pages per set: 1
Laminate: yes

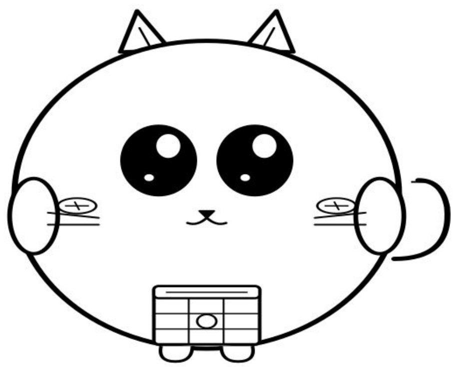
USE IN CLASS
Quantity: 1 puppet
Lessons: 1, 2, 3, 4, 5
Prep: ~12 min

TEACHER PREP
1. Print 1 copy.
2. Cut around silhouette and tab.
3. Laminate.
4. Glue popsicle stick.

# Cal Puppet

(char_cal_puppet)

Paper puppet of Cal the Calendar Cat for teacher use in Lessons 4-5.



Cal the Calendar Cat

PRINT SPECS
Size: letter portrait
Colour: bw
Pages per set: 1
Laminate: yes

USE IN CLASS
Quantity: 1 puppet
Lessons: 4, 5
Prep: ~12 min

TEACHER PREP
1. Print 1 copy.
2. Cut around silhouette and tab.
3. Laminate.
4. Glue popsicle stick.

Name: ____________________ Date: ____________________

# Summative Assessment Rubric

| Expectation | Level 1 (Beginning) | Level 2 (Developing) | Level 3 (Achieving) | Level 4 (Exceeding) |
|---|---|---|---|---|
| **E2.1 use appropriate units of length to estimate, measure, and compare the perimeters of** | Cannot find perimeter. Adds wrong sides. | Finds perimeter with prompts. | Independently measures perimeter and constructs polygons. | Compares perimeters across shapes and explains. |
| **E2.7 compare the areas of two-dimensional shapes by matching, covering, or decomposing and** | Confuses area with perimeter. | Compares areas with prompts. | Independently compares areas by covering or decomposing. | Identifies different shapes with the same area. |
| **E2.9 use square centimetres (cm2) and square metres (m2) to estimate, measure, and compare the** | Cannot use $cm^2$. Wrong unit choice. | Counts $cm^2$ with prompts. | Independently counts $cm^2$ and picks $cm^2$ or $m^2$ appropriately. | Estimates curved-shape area with whole + half squares. |
| **E2.3 use non-standard units appropriately to estimate, measure, and compare capacity, and** | Cannot compare capacity. | Compares with prompts. | Independently measures and compares capacity with non-standard units. | Explains overfill/underfill effects. |
| **E2.6 use analog and digital clocks and timers to tell time in hours, minutes, and seconds** | Confuses hour and minute hands. | Reads analog or digital clock with prompts. | Independently reads analog and digital clocks. | Reads to seconds; converts hours-minutes-seconds. |
| **E2.2 explain the relationships between millimetres, centimetres, metres, and kilometres as** | Cannot pick the right unit for the size. | Picks unit with prompts and benchmarks. | Independently picks mm/cm/m/km using benchmarks. | Converts between units (10 mm = 1 cm; 100 cm = 1 m). |
| **E2.4 compare, estimate, and measure the mass of various objects, using a pan balance and** | Cannot read pan balance. | Reads heavier or lighter with prompts. | Independently compares mass using a pan balance. | Estimates mass before measuring. |
| **E2.5 use various units of different sizes to measure the same attribute of a given item, and** | Thinks different counts mean different attribute. | Recognizes same-attribute with prompts. | Independently shows same attribute under different units. | Articulates inverse relation: bigger unit, fewer counts. |
| **E2.8 use appropriate non-standard units to measure area, and explain the effect that** | Tiles with gaps or overlaps without noticing. | Tiles edge-to-edge with prompts. | Independently tiles edge-to-edge and explains gaps/overlaps. | Quantifies gap/overlap error. |

# Junior Measurer Certificate (G3)

This certificate is proudly presented to:

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For successfully completing the 5-Day G3 Fencing the Farm unit and demonstrating the skills of a G3 Junior Measurer. Roo the Ruler Rabbit congratulates the student.

**By completing this unit, this student has demonstrated:**

- Measuring perimeter and area
- Using $cm^2$ and $m^2$ for area
- Comparing capacity and mass
- Reading analog and digital clocks

# Marketplace Listing

A 5-day Grade 3 unit on measurement framed around Roo the Ruler Rabbit and Cal the Calendar Cat at Farmer Phil's farm.

Price: $9.99 CAD
Pages: ~50
Classroom time: 250 min
Teacher prep: ~220 min

Top tags: Grade 3, math, measurement, perimeter, area, capacity, mass, time

## What's included:

- Unit blueprint (1 file)
- 5 lesson plans (5 files)
- 5 worksheets (5 files)
- Manipulatives plan (1 file)
- Formative + reflection (1 file)
- Assessment suite (1 file)